Multivariate Calculus Lecture Notes
Section 14.4 – Differentiability and Tangent Planes

I. Differentiability for single-variable functions
   a. Remember that a function $f(x)$ is *differentiable* at $x = a$ provided a well-defined tangent line to $f(x)$ exists at a.
   b. Do example with a computer algebra system; use $f(x) = x^2$ and $L(x) = 2x - 1$ near $x = 1$.
   c. This can also be thought of as *local linearity*: If you zoom in close enough around the point $(a, f(a))$, the graph of f becomes indistinguishable from the tangent line.

II. Differentiability for multivariate functions
   a. Intuitively, a multivariate function $f(x, y)$ is *differentiable* at (a, b) if
      i. There is a well-defined tangent plane to $f(x, y)$ that exists at (a, b).
      ii. You zoom in close enough around the point $(a, b, f(a, b))$ the graph of f becomes indistinguishable from the tangent plane.
   b. Show idea with folder and basketball; verify by using a computer algebra system to zoom in on a part of a sphere.
   c. Theorem: If $f_x(x, y)$ and $f_y(x, y)$ exist and are continuous on an open disk D then $f(x, y)$ is differentiable on D.
      i. An open disk is one that does not include its boundary points:

      ii. See Appendix D for a proof.

III. Tangent planes
   a. How would you find the equation of a plane tangent to $f(x, y)$ at the point (a, b)?
   b. Equation of a tangent plane
      i. An important property of the tangent plane at (a, b) is that it must contain the tangent line in the x direction and the tangent line in the y direction.
      ii. In fact, this property defines the tangent plane.
      iii. The tangent line in the x direction
         1. This line has y fixed at b.
         2. So the line equation is $z = f(a, b) + f_x(a, b)(x - a)$.
      iv. Similarly, the tangent line in the y direction is $z = f(a, b) + f_y(a, b)(y - b)$.
      v. Putting these together to obtain a single equation that has both of these as special cases yields the tangent plane equation:

$$L(x, y) = z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

      vi. This has the correct form of a plane equation because it has $f_x(a, b)(x - a) + f_y(a, b)(y - b) - (z - f(a, b)) = 0$.
      vii. This is sometimes called the *linearization* of f at (a, b).
   c. Example: Find the equation of the plane tangent to the graph of $z = y - x^2$ at the point $(4, 3)$.
      i. Graphically:

ii. $f_x(x,y) = -2x$, so $f_x(4,3) = -8$
iii. $f_y(x,y) = 1$, so $f_y(4,3) = 1$
iv. $z(4,3) = -13$.
v. Thus the tangent plane is given by $z = -13 - 8(x-4) + (y-3)$.

IV. Approximating differentiable functions
a. If f is differentiable at (a,b) then $f(x,y) \approx L(x,y)$ near (a,b), as the tangent plane makes a good approximation of f near (a,b).
b. Alternative ways of expressing this
i. Letting $\Delta x = x - a$ and $\Delta y = y - b$, we get

$$f(x,y) = f(a + \Delta x, b + \Delta y) \approx f(a,b) + f_x(a,b)\Delta x + f_y(a,b)\Delta y.$$

ii. Letting $\Delta f = f(x,y) - f(a,b)$, we get

$$\Delta f \approx f_x(a,b)\Delta x + f_y(a,b)\Delta y.$$

iii. Differential form
1. This is $\Delta f \approx df$, where $df = \dfrac{\partial f}{\partial x}dx + \dfrac{\partial f}{\partial y}dy$.
2. Here, we take $dx = \Delta x$ and $dy = \Delta y$ when we use this formula in practice.

c. The radius and height of a circular cone are measured as 10 cm and 25 cm, respectively, with a possible error in measurement of as much as 0.1 cm in each. Use differentials to estimate the volume of a cone with radius 10.1 cm and height 24.9 cm.
i. $V = \dfrac{1}{3}\pi r^2 h$
ii. Thus $dV = \dfrac{2}{3}\pi r h dr + \dfrac{1}{3}\pi r^2 dh$.
iii. For the values of r and h given we have

$$dV = \frac{2}{3}\pi(10)(25)dr + \frac{1}{3}\pi(10)^2 dh = \frac{500}{3}\pi dr + \frac{100}{3}\pi dh.$$

- Thus the volume is more sensitive to changes or errors in r than in h when $r \approx 10$ and $h \approx 25$.

iv. Substituting 0.1 for dx and -0.1 for dy, we have

$$\Delta V \approx dV = \frac{500}{3}\pi(0.1) + \frac{100\pi}{3}(-0.1) = \frac{40\pi}{3} \approx 41.9 \text{ cm}^3.$$

v. The measured volume of the cone is $V = \pi(10)^2(25)/3 = 2618.0$ cm^3.
vi. Thus $V(10.1, 24.9) \approx 2618.0 + 41.9 = 2651.9$ cm^3.

Homework: Section 14.4 #3, 6, 8, 11, 16, 20, 26, 32, 37.